HONDA WB30XH Self-priming Centrifugal Pumpset

Test Report No.	:	2013 - 161
Date of Test	:	December 6, 2013
Test Valid Until	:	December 6, 2018
Brand	:	HONDA
Model	:	WB30XH
Serial No.	:	JH1G8F
Type and Size	:	Single-end suction, 80 x 80 mm, Self-Priming Centrifugal pumpset
Manufacturer	:	Made in China
Test Requested by	:	HONDA PHILIPPINES, INC. First Philippine Industrial Park (FPIP), Tanauan City, Batangas
Machine Classification	:	Commercial

SUMMARY

The HONDA WB30XH Self-priming Centrifugal Pumpset was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump were mounted on a steel frame.

The Maximum Total Head (H) was 30.3 m at zero discharge condition. The shaft speed of the pumpset at this point was 3935 rpm, with Engine Fuel Consumption Rate of 1.25 L/h. The Maximum Discharge Capacity of 15.27 L/s occurred at 4.74 m Total Head with Fuel Consumption Rate of 1.55 L/h. The Maximum Fuel Consumption was 1.57 L/h at 9.30 m Total Head. The shaft speed at this point was 3725 rpm. The Maximum System Efficiency of 11.2% occurred at 11.6 L/s Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 7.50 m. The Net Positive Suction Head Available at this point was 2.30 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pumpset performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pumpset was mounted on the laboratory test set-up shown in **Figure 1.**

The test was conducted on December 6, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna.

Figure 1. HONDA WB30XH Pumpset on the test set-up

METHOD OF TEST

The pumpset was tested following the "Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test".

DESCRIPTION OF THE PUMP

The HONDA WB30XH Self-priming Centrifugal Pumpset shown in **Figure 2**, was a single-end suction, 75 mm x 75 mm, centrifugal pump close-coupled to a 3.6 kW (4.8 Hp) HONDA gasoline engine. It was mounted on a steel base. The suction inlet and discharge outlet were threaded and designed for a 75 mm diameter pipe. The impeller was a semi-open type made of cast iron material connected to the pump shaft through a thread. It was housed in a cast iron double volute casing enclosed in an aluminum chamber. The priming inlet was located on top of the pump beside the discharge outlet. The suction inlet was provided with a swing type one-way valve to retain the sufficient amount of liquid inside the chamber necessary to prime the pump. Located on top of the pump was the discharge outlet. The stuffing box used a mechanical seal.

The primemover, a 3.6 kW HONDA WB30XH gasoline engine was a four-stroke, OHV air-cooled, single inclined cylinder gasoline engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized a dual element air cleaner moistened with oil. Lubrication system was force-feed lubricating type. Cooling was done by air forced by a fan/flywheel encased in a metal shroud. It was started by a rope-recoil system. A muffler was used in the exhaust system. **Table 1** shows the detailed specifications of the pump.

Figure 2. HONDA WB30XH Self-priming Centrifugal Pumpset

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the HONDA WB30XH Self-priming Centrifugal Pumpset

1.	Brand		:	HONDA
2.	Model		:	WB30XH
3.	Serial No.		:	JH1G8F
4.	Make		:	China
5.	Type		:	Single-end suction, Self-priming centrifugal pumpset
6.	Size (mm)		:	75 x 75
7.	Overall Dimension and Weight (Including mounting frame and engine)			
	7.1	Length (mm)	:	510
	7.2	Width (mm)	:	309
	7.3	Height (mm)	:	437
	7.4	Weight (kg)	:	38
8.	Impeller			
	8.1	Type	:	Semi-open
	8.2	Diameter (mm)	:	116.5
	8.3	Width (mm)	:	27.0
	8.4	No. of vanes	:	3
	8.5	Material	:	Cast iron
9.	Suction Inlet			
	9.1	Type	:	Threaded flange
	9.2	Diameter (mm)	:	75
	9.3	Material	:	Aluminum
10.	Discharge Outlet			
	10.1	Type	:	Threaded flange
	10.2	Diameter (mm)	:	75
	10.3	Material	:	Aluminum

Table 1. (Cont'n)

11.	Casing		
	11.1 Type	:	Double volute
	11.2 Material	:	Cast Iron
12	Stuffing box	:	Mechanical Seal
13.	Rotation	:	Counter clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	3600
15.	Rated Maximum Discharge Capacity (L/s)	:	18.3
16.	Rated Maximum Total Head (m)	:	28
17.	Primemover (based on Plate Rating)		
	17.1 Brand	:	HONDA
	17.2 Model	:	GX160
	17.3 Serial No.	:	GCAAH-3358133
	17.4 Make	:	China
	17.5 Type	:	Four stroke cycle, OHV air-cooled, single inclined cylinder gasoline engine
	17.6 Rated Output Power (kW)		
	Maximum	:	3.6 @ 3600 rpm
	Continuous	:	No Data
	17.8 Bore x Stroke (mm)	:	No Data
	17.9 Displacement Volume (cc)	:	No Data
	17.10 Cooling System	:	Air-cooled
	17.11 Lubricating System	:	Force-feed
	17.12 Starting System	:	Rope-recoil
	17.13 Air Cleaner	:	Dual element
	17.14 Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pumpset Tested	:	HONDA WB30XH
Date of Test	:	December 6, 2013
Test Condition		
Ambient Temperature (^{o}C)	:	27.3
Atmospheric Pressure (mb)	:	1013
Pumping Water Temperature (^{o}C)	:	28.8

Table 2. Results of Performance Test

PUMP SHAFT SPEED (rpm)	FUEL CONSUMPTION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY (%)
3935	1.25	30.3	0	0	12.2	0
3835	1.27	24.8	1.25	0.30	12.4	2.45
3785	1.35	22.5	4.17	0.92	13.1	7.00
3755	1.42	20.3	6.54	1.30	13.8	9.49
3750	1.48	18.6	8.24	1.50	14.3	10.5
3750	1.54	16.5	9.98	1.62	14.9	10.8
3730	1.54	14.6	11.6	1.67	14.9	11.2
3725	1.56	12.8	12.6	1.59	15.2	10.5
3725	1.57	9.30	14.7	1.34	15.2	8.8
3725	1.56	8.01	15.2	1.19	15.2	7.85
3755	1.55	4.74	15.7	0.73	15.0	4.85

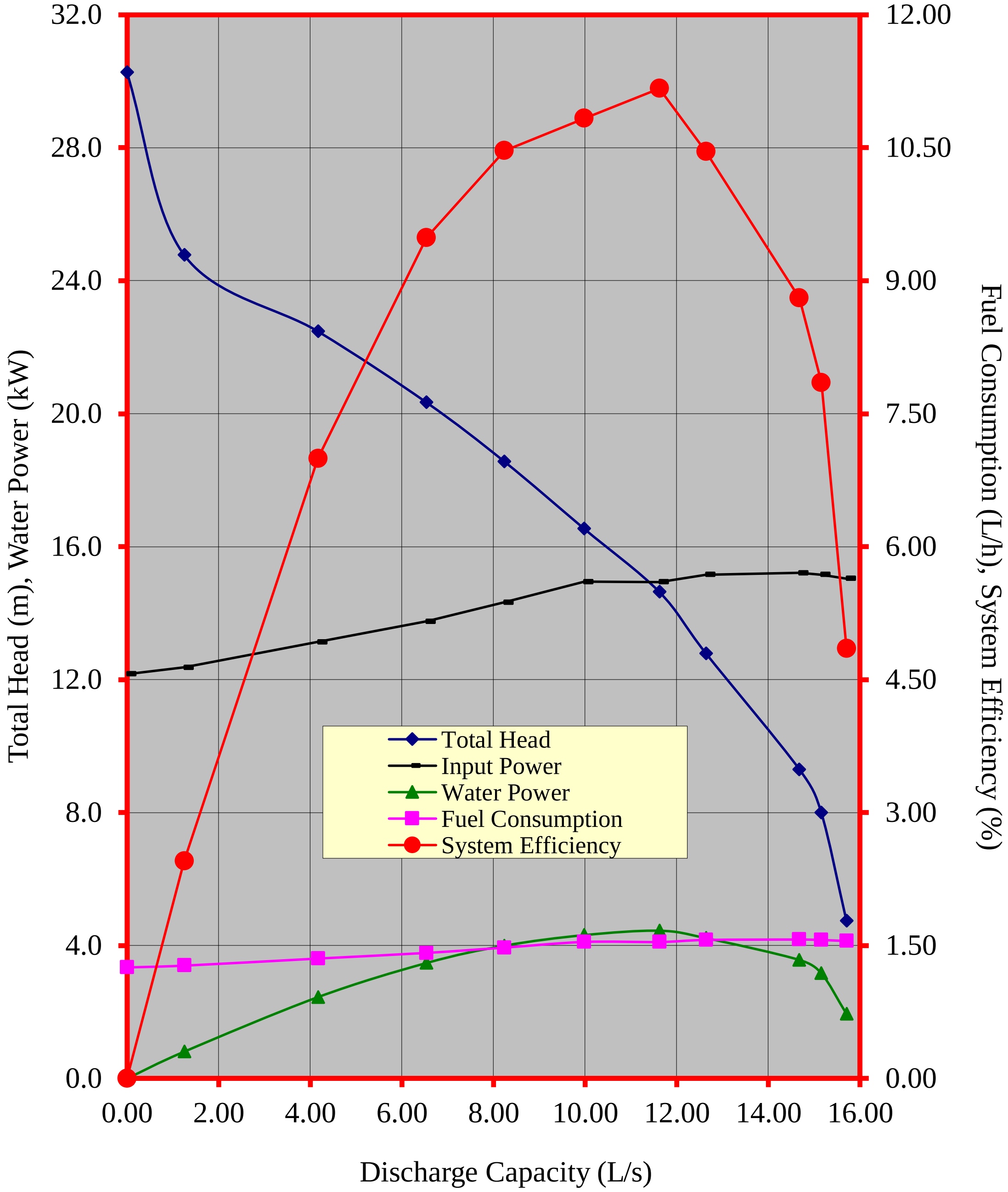

Figure 3. Performance characteristic curves of the HONDA WB30XH self-priming centrifugal pumpset

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pumpset Tested	:	HONDA WB30XH
Date of Test	:	December 6, 2013
Test Condition		
Ambient Temperature (^{o}C)	:	27.6
Atmospheric Pressure (mb)	:	1013
Pumping Water Temperature (^{o}C)	:	29.3

Table 3. Results of Cavitation Test

PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
3755	1.31	11.1	5.11	16.6
3730	0.23	9.56	6.34	16.1
3750	1.94	7.85	7.63	15.2
3880	4.88	4.91	8.59	10.1
3885	6.25	3.54	9.18	6.79
3910	7.50	2.30	9.67	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

Where: $$H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted to meter. Computed by determining the velocity of liquid on the pipe by formula, V = Q/A where V, velocity of liquid inside the pipe in meter per second, Q, Discharge rate of the pump in liters per second, and A, Cross-sectional area of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q, in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.

3. Water Power, WP, is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power, IP, is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determined by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters
P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

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